

Docker Notes

Installing Docker

```
sudo su
yum update -y
yum install docker -y      → This will install into your system
docker --version
service docker status
service docker start      → docker starts
```

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri May 22 09:48:33 2026 from 13.48.4.203
ubuntu@ip-172-31-43-128:~$ sudo su
root@ip-172-31-43-128:/home/ubuntu# apt update
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [136 kB]
Hit:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:4 http://security.ubuntu.com/ubuntu resolute-security InRelease [136 kB]
Get:5 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [90.6 kB]
Get:6 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [9956 B]
Get:7 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [45.5 kB]
Get:8 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [27.4 kB]
Get:9 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64v3 Packages [3360 B]
Get:10 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [772 B]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [2848 B]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [20.6 kB]
Fetched 472 kB in 0s (1974 kB/s)
34 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ip-172-31-43-128:/home/ubuntu# apt install docker.io -y
docker.io is already the newest version (29.1.3-0ubuntu4.1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 34
root@ip-172-31-43-128:/home/ubuntu#
```

Creating a Container

How we can create a particular container onto this machine:

```
docker run -it ubuntu /bin/bash
```

If I want to create another container, run the same above command.

Essential Commands

```
root@ip-172-31-43-128:/home/ubuntu# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS                PORTS                               NAMES
1765e1799858   ubuntu   "/bin/bash"             53 minutes ago Up 50 minutes        0.0.0.0:1234->80/tcp, [::]:1234->80/tcp   cont3
447863dcf2a9   ubuntu   "/bin/bash"             58 minutes ago Exited (127) 58 minutes ago                                cont1
ea122941d724   ubuntu   "/bin/bash"             2 hours ago    Exited (0) 2 hours ago                                conl
c4fcbf2413f0   ubuntu   "/bin/bash"             3 hours ago    Up 3 hours                                     kumarcon2
402275098fe5   ubuntu   "/bin/bash"             4 hours ago    Up 4 hours                                     kumarcon
root@ip-172-31-43-128:/home/ubuntu#
```

1. Check all created containers:

```
| docker ps -a
```

2. Check images in container:

```
| docker images → to see images present in local registry
```

3. For starting a particular container:

```
| docker start container_name
```

4. For entering into container:

```
| docker attach name
```

5. ls – to show/see all files:

```
| ls → press Enter
```

6. Get into tmp folder by typing:

```
| cd tmp
```

7. Create few files:

```
| touch file1 file2 file3
```

```
root@ip-172-31-43-128:/home/ubuntu# docker run -it --name cont4 /bin/bash
docker: invalid reference format

Run 'docker run --help' for more information
root@ip-172-31-43-128:/home/ubuntu# docker run -it --name cont4 ubuntu /bin/bash
root@8a904e019a5d:/# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@8a904e019a5d:/# cd tmp
root@8a904e019a5d:/tmp# touch file1 file2 file3
root@8a904e019a5d:/tmp# ls
file1 file2 file3
root@8a904e019a5d:/tmp#
```

Coming Out of Container

- Press Ctrl + P → saves process
- Press Ctrl + Q → quit process

Dockerfile

Text file consisting of instructions & data.

```
FROM ubuntu
VOLUME ["/myVolume"]
WORKDIR /tmp
RUN echo "hello kumar " > /tmp/testfile
COPY testfile /tmp
```

```
| vi Dockerfile
```

Inside the Dockerfile:

```
FROM ubuntu
WORKDIR /tmp
RUN echo "hello learner" > /tmp/testfile
COPY testfile /tmp
```

Creating a Volume in Dockerfile

```
FROM ubuntu
VOLUME ["/myvolume"]
WORKDIR /tmp
RUN echo "hello learner" > /tmp/testfile
COPY test /tmp
```

Creating New Container with Name

```
| docker run -it --name kumarcon ubuntu /bin/bash
```

```
FROM ubuntu
VOLUME ["/myVolume"]
WORKDIR /tmp
RUN echo "hello learner" > /tmp/testfile
COPY test
```

Volume Commands

To see list of volumes:

```
| docker volume ls
```

To create a docker volume:

```
| docker volume create kumarvol
```

To create a container of which a directory named logs will be mapped with kumarvol:

```
| docker run -it -v kumarvol:/logs --name con1 ubuntu /bin/bash
```

To see where volume is created:

```
| docker inspect volumename
```

From that you'll get mountpoint. Type ls + mountpoint to see files in docker volume:

```
| ls <mountpoint>
```

Starting Apache2 Server

```
Last login: Fri May 22 07:12:55 2026 from 13.48.4.203
ubuntu@ip-172-31-43-128:~$ sudo su
root@ip-172-31-43-128:/home/ubuntu# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
1765e1799858   ubuntu   "/bin/bash"             16 minutes ago Up 13 minutes  0.0.0.0:1234->80/tcp, [::]:1234->80/tcp   cont3
c4fcbf2413f0   ubuntu   "/bin/bash"             3 hours ago   Up 3 hours                               kumarcon2
402275098fe5   ubuntu   "/bin/bash"             3 hours ago   Up 3 hours                               kumarcon
root@ip-172-31-43-128:/home/ubuntu# docker exec -it cont3 /bin/bash
root@1765e1799858:/# apt install apache2
Error: Unable to locate package apache2
root@1765e1799858:/# apt-get update
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [136 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [136 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [102 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [51.7 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages [352 kB]
0% [6 Packages store 82.9 kB] [7 Packages 311 kB/352 kB 88%] [Waiti                                     Get:8
http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [216 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages
20.1 MB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/restricted amd64 Packages [189 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages [1874 kB]
```

To get update downloads:

```
| apt-get update
```

To install apache2 server in our docker container (cont3):

```
| apt-get install apache2 -y
```

Then start the server:

```
| service apache2 start
```

```
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of start.
Processing triggers for libc-bin (2.43-2ubuntu2) ...
Processing triggers for ca-certificates (20260223) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
root@1765e1799858:/# service apache2 start
 * Starting Apache httpd web server apache2
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.17.0.4. Set the 'ServerName' directive globally to suppress this message
 *
root@1765e1799858:/# service apache2 status
 * apache2 is running
root@1765e1799858:/# cd /var/www/html
bash: cd /var/www/html: No such file or directory
root@1765e1799858:/# exit
exit
root@ip-172-31-43-128:/home/ubuntu# docker exec -it cont3 /bin/bash
root@1765e1799858:/# cd /var/www/html
root@1765e1799858:/var/www/html# echo < "hello" > index.html
bash: hello: No such file or directory
root@1765e1799858:/var/www/html# echo < "hello" > index.html
```

Note: You can clone any projects from GitHub by cloning from there.

Note: You have to install Git in your container (cont3).

At last you'll get an IP address which is public.

Docker Notes | Nandan Kumar